

Retail Customer Purchase & Sales Analysis

1. Project Overview

Project Title

Retail Customer Purchase & Sales Analysis

Project Type

Customer Analytics | Retail Analytics | Exploratory Data Analysis | SQL Analysis

Objective

The objective of this project was to analyze customer purchasing behavior and sales patterns using a retail customer dataset. The analysis focuses on understanding:

- Customer spending behavior
- Gender-wise revenue contribution
- Product performance
- Customer loyalty and purchase history
- Discount and promotional behavior
- Shipping preferences
- Subscription behavior
- Age-group revenue contribution
- Category-level purchasing patterns
- Seasonal sales and revenue
- Geographic order distribution
- Product review performance

The project combines **Python for data cleaning and feature engineering** with **MySQL for structured business analysis using SQL**.

2. Dataset Description

The dataset contains **3,900 customer purchase records** and initially contains **18 columns**.

Column	Description
Customer ID	Unique identifier for each customer
Age	Customer age

Column	Description
Gender	Customer gender
Item Purchased	Product purchased
Category	Product category
Purchase Amount (USD)	Amount spent on the purchase
Location	Customer location
Size	Product size
Color	Product color
Season	Season of purchase
Review Rating	Customer rating for the purchased item
Subscription Status	Whether the customer is subscribed
Shipping Type	Shipping method selected
Discount Applied	Whether a discount was applied
Promo Code Used	Whether a promotional code was used
Previous Purchases	Number of previous purchases
Payment Method	Payment method
Frequency of Purchases	Customer purchase frequency

Dataset Size

- **Rows:** 3,900
- **Columns:** 18
- **Purchase Amount:** USD
- **Age range:** 18–70
- **Purchase amount range:** USD 20–100
- **Review rating range:** 2.5–5.0

3. Tools and Technologies

The following tools were used:

Python

Used for:

- Data inspection

- Data cleaning
- Missing-value treatment
- Column-name standardization
- Feature engineering
- Frequency transformation

MySQL

Used for:

- Data storage
- SQL querying
- Aggregation
- Customer segmentation
- Ranking products
- Revenue analysis
- Category-level analysis
- Window functions and CTEs

Key Python Libraries

- Pandas
 - NumPy
-

4. Data Understanding and Initial Inspection

The dataset was initially inspected to understand:

- Number of observations
- Data types
- Missing values
- Numerical distributions
- Categorical variables

The initial structure showed:

- 3,900 records
- 4 integer columns
- 1 floating-point column
- 13 string columns

Missing Values

The primary missing-value issue was observed in **Review Rating**, which contained:

3,863 non-null values out of 3,900

Therefore:

$3,900 - 3,863 = 37$ missing review ratings

The missing review ratings were handled using **median-based imputation at the category level**, rather than simply replacing every missing value with one global median.

This approach is preferable because review-rating behavior can differ between product categories.

5. Data Cleaning

Several preprocessing steps were performed before analysis.

5.1 Column Name Standardization

Column names were cleaned to make them easier to use in Python and SQL.

For example:

Purchase Amount (USD)

was converted to:

`purchase_amount`

Other column names were standardized by:

- Removing unnecessary spaces
 - Converting names to lowercase
 - Replacing spaces with underscores/removing unwanted spaces
-

5.2 Missing-Value Treatment

Missing values were identified and handled before performing the analysis.

For the review-rating field, median-based imputation was used.

This prevented missing values from affecting:

- Average rating calculations
 - Customer review analysis
 - Product rating comparisons
-

6. Feature Engineering

6.1 Age Group

Customers were divided into four age groups using quartile-based segmentation with `pd.qcut()`.

The groups were:

1. Young Adult
2. Adult
3. Middle Aged
4. Senior Citizen

The use of quartiles ensures that the age groups are based on the distribution of customers rather than arbitrary age boundaries.

This feature was later used to analyze revenue contribution by customer age group.

6.2 Purchase Frequency Conversion

The categorical purchase-frequency values were converted into approximate numbers of days between purchases.

Frequency	Approx. Days
Weekly	7
Fortnightly	14
Bi-Weekly	14
Monthly	30
Every 3 Months	90
Quarterly	91
Annually	365

This transformation converts a categorical business concept into a numerical variable that could potentially be used for:

- Customer frequency analysis
 - Customer segmentation
 - Retention analysis
 - Purchase-cycle analysis
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7. Data Loading into MySQL

After preprocessing, the cleaned dataset was loaded into MySQL.

A database named:

retail_analysis

was created.

The cleaned dataset was stored in:

customer_analysis

The SQL environment was then used to perform business-oriented analysis.

8. Business Questions and SQL Analysis

Question 1: What is the total revenue generated by male and female customers?

The following aggregation was performed:

```
SELECT
    gender,
    SUM(purchase_amount) AS total_amount
FROM customer_analysis
GROUP BY gender;
```

Result

	gender	total_amount
▶	Male	157890
	Female	75191

Key Finding

Male customers generated substantially higher revenue than female customers in this dataset.

Total revenue across both groups:

\$233,081

Male customers contributed approximately **67.7%** of the total revenue, while female customers contributed approximately **32.3%**.

9. Promotional Customers With Above-Average Purchases

Business Question

Which customers used a discount but still made purchases equal to or greater than the overall average purchase amount?

```
SELECT
    customer_id,
    purchase_amount
```

```

FROM customer_analysis
WHERE discount_applied = 'Yes'
AND purchase_amount >= (
    SELECT AVG(purchase_amount)
    FROM customer_analysis
);

```

	customer_id	purchase_amount
▶	2	64
	3	73
	4	90
	7	85
	9	97
	12	68
	13	72
	16	81

Analytical Purpose

This analysis identifies customers who:

- Received a discount
- Still spent at or above the overall average

Business Value

These customers may represent an important segment because the discount did not prevent them from generating relatively high transaction value.

However, this analysis **does not prove that the discount caused the higher spending**. A controlled comparison between discounted and non-discounted customers would be required to establish that relationship.

10. Top 5 Products by Average Review Rating

```

SELECT
    item_purchased,
    ROUND(AVG(review_rating),2) AS avg_rating
FROM customer_analysis
GROUP BY item_purchased
ORDER BY avg_rating DESC
LIMIT 5;

```

	item_purchased	avg_rating
►	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.8
	Handbag	3.78

Objective

Identify products receiving the highest average customer ratings.

Business Application

High-rated products may be candidates for:

- Greater visibility
- Featured-product placement
- Cross-selling
- Promotional campaigns
- Customer recommendation systems

However, average rating should be evaluated together with **number of reviews**. A product with a 5.0 rating from a very small number of customers may be less reliable than a product with a 4.7 rating from hundreds of customers.

11. Standard vs Express Shipping

SELECT

shipping_type,

ROUND(AVG(purchase_amount),2) AS avg_purchase

FROM customer_analysis

WHERE shipping_type IN ('Standard','Express')

GROUP BY shipping_type;

	shipping_type	avg_purchase
►	Express	60.48
	Standard	58.46

Objective

Compare average purchase value between customers selecting Standard and Express shipping.

Business Question

Do customers choosing faster shipping also tend to make higher-value purchases?

Business Application

If Express customers have a significantly higher average purchase amount, the company could investigate whether premium shipping is associated with high-value customers.

This could support:

- Shipping upsell strategies
 - Premium customer segmentation
 - Delivery-pricing decisions
-

12. Subscription Analysis

SELECT

```
subscription_status,  
COUNT(customer_id) AS total_customer,  
ROUND(AVG(purchase_amount),2) AS total_spend,  
COUNT(review_rating) AS total_reviews
```

FROM customer_analysis

GROUP BY subscription_status

ORDER BY total_customer DESC;

	subscription_status	total_customer	total_spend	total_reviews
▶	No	2847	59.87	2847
	Yes	1053	59.49	1053

Objective

Compare subscribed and non-subscribed customers using:

- Customer count
- Average purchase amount
- Number of recorded reviews

Business Question

Do subscribed customers spend more than non-subscribed customers?

13. Products With the Highest Discount Usage

SELECT

```
item_purchased,  
ROUND(  
  SUM(  
    CASE  
      WHEN discount_applied = 'Yes' THEN 1  
      ELSE 0
```

```

        END
    ) * 100.0 / COUNT(*),
    2
) AS purchased_percentage
FROM customer_analysis
GROUP BY item_purchased
ORDER BY purchased_percentage DESC
LIMIT 5;

```

	item_purchased	purchased_percentage
▶	Hat	50.00
	Sneakers	49.66
	Coat	49.07
	Sweater	48.17
	Pants	47.37

Objective

Identify the five products with the highest percentage of purchases made using discounts.

Business Value

This can help identify products that are:

- Frequently promoted
- More discount-dependent
- Potentially price-sensitive
- Strong candidates for promotional campaigns

14. Customer Segmentation

Customers were segmented based on their previous purchase history.

Segmentation Logic

Previous Purchases Segment

1	New
2–10	Returning
>10	Loyal

SQL logic:

```

with cte as (
select customer_id,previous_purchases,
(case

```

```

when previous_purchases=1 then 'New'
when previous_purchases between 2 and 10 then 'Returning'
else 'Loyal' end ) as customer_segment
from customer_analysis)

```

```

select customer_segment , count(*) as number_of_people
from cte

```

```

group by 1;

```

	customer_segment	number_of_people
▶	Loyal	3116
	Returning	701
	New	83

Business Purpose

This segmentation helps the business understand:

- Customer retention
- Repeat purchasing
- Loyal customer contribution
- Opportunities for retention campaigns

Important Definition Note

Calling customers with exactly **1 previous purchase "New"** is a business definition rather than a universal definition. They have technically already purchased before.

15. Top 3 Products Within Each Category

A CTE and ROW_NUMBER() were used to rank products within each category.

```

WITH cte1 AS (
    SELECT
        category,
        item_purchased,
        COUNT(*) AS purchase_count
    FROM customer_analysis
    GROUP BY category, item_purchased
),
cte2 AS (

```

```

SELECT
    category,
    item_purchased,
    purchase_count,
    ROW_NUMBER() OVER (
        PARTITION BY category
        ORDER BY purchase_count DESC
    ) AS rnk
FROM cte1
)
SELECT
    category,
    item_purchased,
    purchase_count
FROM cte2
WHERE rnk <= 3;

```

	category	item_purchased	purchase_count
►	Accessories	Jewelry	171
	Accessories	Sunglasses	161
	Accessories	Belt	161
	Clothing	Blouse	171
	Clothing	Pants	171
	Clothing	Shirt	169
	Footwear	Sandals	160
	Footwear	Shoes	150
	Footwear	Sneakers	145
	Outerwear	Jacket	163
	Outerwear	Coat	161

Objective

Identify the three most frequently purchased products within every product category.

Business Application

This analysis can support:

- Inventory planning
- Product recommendations
- Category merchandising
- Cross-selling
- Product placement

The use of PARTITION BY category ensures that products are ranked independently within each category.

16. Repeat Buyers and Subscription Status

SELECT

subscription_status,

COUNT(*) AS customers

FROM customer_analysis

WHERE previous_purchases > 5

GROUP BY subscription_status;

	subscription_status	customers
▶	Yes	958
	No	2518

Objective

Examine subscription status among customers who have made more than five previous purchases.

Business Question

Among repeat/experienced customers, how are customers distributed between subscribers and non-subscribers?

"What proportion of highly repeat customers are subscribers versus non-subscribers?"

17. Revenue Contribution by Age Group

SELECT

age_group,

count(customer_id) as total_customer,

SUM(purchase_amount) AS total_revenue

FROM customer_analysis

GROUP BY age_group

ORDER BY total_revenue DESC;

	age_group	total_customer	total_revenue
▶	Young Adult	1028	62143
	Middle Aged	986	59197
	Senior Citizen	944	55763
	Adult	942	55978

Objective

Measure the contribution of each age group to total revenue.

Business Application

This helps identify the customer age segments generating the greatest revenue.

The result can be used to support:

- Targeted marketing
- Product positioning
- Customer segmentation
- Campaign personalization

18. Geographic Analysis

The original query was:

```
SELECT
```

```
    location,
```

```
    COUNT(customer_id) AS total_orders,
```

```
    sum(purchase_amount)as total_revenue
```

```
FROM customer_analysis
```

```
GROUP BY location
```

```
ORDER BY total_orders DESC;
```

	location	total_orders	total_revenue
►	Montana	96	5784
	California	95	5605
	Idaho	93	5587
	Illinois	92	5617
	Alabama	89	5261
	Minnesota	88	4977
	Nevada	87	5514
	Nebraska	87	5172
	New York	87	5257
	Delaware	86	4758
	Maryland	86	4795
	Vermont	85	4860
	Louisiana	84	4848
	North Da...	83	5220
	New Mexico	81	5014
	West Virg...	81	5174
	Missouri	81	4691
	Mississippi	80	4883
	Indiana	79	4655
	Arkansas	79	4828
	Kentucky	79	4402
	Georgia	79	4645
	North Ca...	78	4742

Objective

Identify locations with the highest number of orders.

Business Application

This can support:

- Regional marketing
- Inventory planning
- Logistics optimization
- Regional campaign targeting

19. Average Purchase Amount by Category

SELECT

category,

ROUND(AVG(purchase_amount),2) AS avg_purchase_amount

FROM customer_analysis

GROUP BY category

ORDER BY avg_purchase_amount DESC;

	category	avg_purchase_amount	avg_orders
►	Footwear	60.26	599
	Clothing	60.03	1737
	Accessories	59.84	1240
	Outerwear	57.17	324

Objective

Determine which product categories have the highest average transaction value.

Business Application

This can reveal categories associated with higher customer spending and help inform:

- Product assortment
- Pricing strategy
- Promotional strategy
- Cross-selling opportunities

20. Median Purchase Amount by Category

A window-function approach was used to calculate the median purchase amount for each category.

WITH ranked AS (

```

SELECT
    category,
    purchase_amount,
    ROW_NUMBER() OVER (
        PARTITION BY category
        ORDER BY purchase_amount
    ) AS rn,
    COUNT(*) OVER (
        PARTITION BY category
    ) AS total_count
FROM customer_analysis
)
SELECT
    category,
    AVG(purchase_amount) AS median_purchase_amount
FROM ranked
WHERE rn IN (
    FLOOR((total_count + 1) / 2),
    CEIL((total_count + 1) / 2)
)
GROUP BY category;

```

	category	median_purchase_amount
►	Accessories	60.0000
	Clothing	60.0000
	Footwear	60.0000
	Outerwear	54.5000

Why Median Was Used

Average purchase value can be influenced by unusually high or low purchases.

The median represents the middle purchase and is therefore more robust against outliers.

Analytical Value

Comparing:

Average Purchase Amount vs Median Purchase Amount

can help identify whether a category's spending distribution is skewed.

For example:

- Average $\approx$ Median $\rightarrow$ relatively balanced distribution
- Average $>$ Median $\rightarrow$ potentially right-skewed due to high-value purchases
- Average $<$ Median $\rightarrow$ potentially left-skewed

21. Seasonal Sales Analysis

```
SELECT
    season,
    COUNT(customer_id) AS total_sales
FROM customer_analysis
GROUP BY season
ORDER BY total_sales DESC;
```

	season	total_sales
►	Spring	999
	Fall	975
	Winter	971
	Summer	955

Objective

Determine the number of purchases made during each season.

This helps identify periods with higher transaction volume.

22. Seasonal Revenue Analysis

```
SELECT
    season,
    SUM(purchase_amount) AS total_price
FROM customer_analysis
GROUP BY season
ORDER BY total_price DESC;
```

	season	total_price
►	Fall	60018
	Spring	58679
	Winter	58607
	Summer	55777

Objective

Measure total revenue generated during each season.

Important Distinction

Two different seasonal metrics were analyzed:

Sales volume:

COUNT(customer_id)

Revenue:

SUM(purchase_amount)

A season can have:

- High order volume but lower revenue
- Lower order volume but higher revenue

Therefore, both metrics should be considered when evaluating seasonal performance.

23. Key Findings

Based on the results currently available, the strongest confirmed finding is:

1. Male customers generated higher total revenue

Male customers generated:

\$157,890

compared with:

\$75,191

from female customers.

Male customers therefore contributed approximately **67.7% of total revenue** in the dataset.

However, further analysis should compare customer counts and average purchase amount by gender before concluding that male customers have higher individual customer value.

24. Analytical Framework

The project followed the following analytical workflow:

Raw Dataset

↓

Data Inspection

↓

Data Cleaning

- Standardized columns
- Identified missing values
- Handled missing review ratings
- Cleaned categorical fields



Feature Engineering

- Age groups
- Purchase-frequency mapping



MySQL Database



Business Questions

- Revenue
- Products
- Customers
- Discounts
- Subscriptions
- Shipping
- Categories
- Seasonality
- Geography



SQL Analysis

- Aggregations
- CTEs
- CASE statements
- Window functions
- Ranking
- Median calculations



Business Insights



Recommendations

25. Business Recommendations

Based on the analytical framework, the following recommendations can be considered.

1. Investigate gender-based revenue differences

Male customers generated substantially higher total revenue. The next step should be to compare:

- Customer count
- Average purchase amount
- Purchase frequency
- Product-category preferences

This will determine whether the difference comes from customer volume or higher individual spending.

2. Optimize discount strategy

Products with high discount usage should be evaluated against:

- Revenue
- Average purchase value
- Customer retention
- Profit margin

High discount usage does not necessarily mean a successful promotion.

3. Focus on loyal customers

Customers with a high number of previous purchases can be treated as a valuable retention segment.

Potential strategies include:

- Loyalty rewards
- Personalized offers
- Early product access
- Cross-selling
- Subscription incentives

4. Use category-level product rankings for inventory planning

The top products within each category can help prioritize:

- Inventory availability
- Product placement
- Recommendations
- Promotional campaigns

5. Analyze seasonal demand

High-volume and high-revenue seasons should be compared separately.

This can support:

- Inventory planning
 - Marketing campaigns
 - Staffing
 - Shipping capacity planning
-

27. Conclusion

This project demonstrates an end-to-end retail analytics workflow, beginning with raw customer purchase data and progressing through data cleaning, feature engineering, database loading, SQL analysis, and business interpretation.

The analysis examined customer demographics, purchasing behavior, product performance, discounts, subscription status, customer loyalty, shipping preferences, geographic distribution, category performance, and seasonal trends.

The project also demonstrates practical SQL skills including:

- GROUP BY
- Aggregations
- Subqueries
- CASE WHEN
- CTEs
- ROW_NUMBER()
- PARTITION BY
- Window functions
- Ranking
- Median calculation

The analysis provides a foundation for further work in customer segmentation, retention analysis, product performance, and revenue optimization.

28. Skills Demonstrated

Technical Skills

- Python
- Pandas
- NumPy
- MySQL
- SQL
- Data Cleaning

- Feature Engineering
- CTEs
- Window Functions
- Data Aggregation
- Statistical Summaries

Analytical Skills

- Business-question formulation
- Customer segmentation
- Revenue analysis
- Product analysis
- Promotional analysis
- Customer behavior analysis
- Seasonal analysis
- Geographic analysis
- KPI interpretation
- Business recommendation

Business Concepts

- Revenue
- Average Order Value
- Customer segmentation
- Customer loyalty
- Repeat purchasing
- Promotional effectiveness
- Product performance
- Customer reviews
- Seasonal demand
- Shipping behavior